Problem Set 2

Graded

Student

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Total Points

95 / 100 pts

Question 1

Question 1

40 / 40 pts

- ✔ **– 0 pts** Correct
- ✔ **– 0 pts** Follow Homework format

– 1 pt Incorrect value used

– 1 pt Incorrect value used

– 1 pt Missing/Incorrect units

– 1 pt Missing/Incorrect units

– 1 pt Calculation error

– 1 pt Incorrect equation used

– 1 pt Incorrect equation used

– 1 pt Incorrect final answer

– 1 pt Incorrect final answer

– 10 pts Missing part C

– 40 pts Problem not attempted

Question 2

Question 2

35 / 40 pts

- **0 pts** Correct

✔ **0 pts** Follow homework format

- **20 pts** Incorrect a)
- **10 pts** Major error in a)
- **10 pts** Incorrect b)
- **10 pts** Incorrect c)
- **2 pts** c) Error in strategy
- **5 pts** c) 2 Errors in strategies
- **1 pt** Incorrect/missing units

✔ **5 pts** major error in energy balance

1 need to do control vol at surface and then at x=0. your method is only considering conduction here

- **2 pts** calculation error

Question 3

Question 3

20 / 20 pts

- **1 pt** Missing the negative sign for the k
- **10 pts** Incorrect b: claimed right side has larger heat flux
- **1 pt** Select pages in the future for full credit.

✔ **0 pts** Perfect

- **10 pts** Incorrect direction for a
- **5 pts** Partially incorrect a: either direction for arrows is not fully correct or no arrows drawn at all
- **20 pts** no submission

Question assigned to the following page: 1

Heat transfer HW2

1. KNOWN: $R = 0.5\,m$, $H = 1\,m$, $T_w = 50°C$, $T_s = 50°C$, $T_\infty = 20°C$

$h = 15\,W/(m^2 \cdot K)$, (b) $L = 2\,cm$, $k = 0.1\,W/(m \cdot K)$

FIND: (a) Total electrical heating power q to maintain $T_w = 50°C$

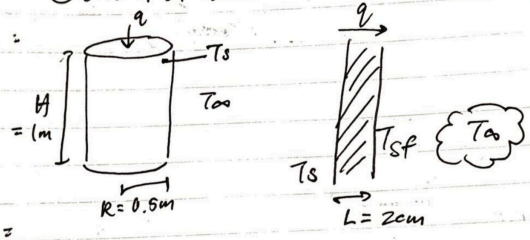
(b) Total electrical heating power q w/ insulation

(c) Role of thermal insulation in energy saving.

ASSUMPTIONS: ① Steady state ② Wall of water heater thin & high thermal conductivity

③ Bottom of heater well insulated ④ Thermal radiation negligible

SCHEMATIC:



ANALYSIS:

(a) q = heat loss to environment

$q = q_{cond} + q_{conv}$. Since exterior wall is the same temperature as the water, there is no heat loss through conduction $q_{cond} = 0$.

$q = q_{conv}$.

$q = hA(T_s - T_\infty)$ $\quad A = 2\pi RH + \pi R^2$

$= (15)(3.927)(50-20)$ $\quad = 3.927\,m^2$

$= \boxed{1767\,W}$

(b) Assuming 1D heat conduction

Conduction from exterior wall to exterior of foam:

$q_{cond} = kA\frac{dT}{dx}$ $\quad A = 2\pi(R+L)(H+L) + \pi(R+L)^2$

$= kA\frac{(T_s - T_{sf})}{L}$ $\quad = 4.182\,m^2$

Convection from exterior of foam into surroundings:

$q_{conv} = hA(T_{sf} - T_\infty)$ $\quad q_{cond} = \boxed{470.49\,W}$

$q_{cond} = q_{conv}$

$kA\frac{(T_s - T_{sf})}{L} = hA(T_{sf} - T_\infty)$

$(0.1)\frac{(50 - T_{sf})}{0.02} = (15)(T_{sf} - 20)$

$T_{sf} = 27.5°C$

Questions assigned to the following page: 1 and 2

(c) The power needed for part (b) is much lower than that in part (a). Due to the low thermal conductivity of the foam, shielding the water heater from heat loss, less energy is needed to make up for it.

2. KNOWN: $L = 10\,cm$, $k = 1\,W/(m\,K)$, $T_\infty = 20°C$, $h = 100\,W/(m^2 \cdot K)$

(a) $q'' = 1000\,W/m^2$

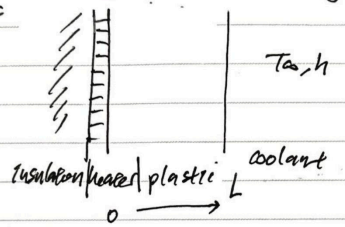
FIND: (a) Temp. of interior surface & exterior surface

(b) Max. heat flux without overheating

(c) Three strategies for higher flux without exceeding 200°C

ASSUMPTIONS: ① Steady state ② Infinitely thin ③ Neglect thermal radiation

SCHEMATIC:



ANALYSIS: (a) $q'' = 1000\,W/m^2$

~~$T_{x=0} = T_{heater}$~~

$q'' = q''_{cond.}$ and $q''_{cond} = q''_{coolant}$

$1000 = h(T_L - T_\infty)$

$1000 = 100(T_L - 20)$

$\boxed{T_L = 30°C}$

$1000 = k\left(\frac{T_0 - T_L}{L}\right)$

$1000 = (1)\left(\frac{T_0 - 30}{0.1}\right)$

$\boxed{T_0 = 130°C}$

(b) $T_{max} = 200°C$

$q''_{max} = k\left(\frac{200 - T_L}{0.1}\right)$

$= (1)\left(\frac{200 - 30}{0.1}\right)$

$q''_{max} = 1700\,W/m^2$

(c) ① $q = k\left(\frac{T_{max} - T_L}{L}\right)$

Use material with higher thermal conductivity

② decrease temperature of coolant

③ decrease thickness of plastic layer.

Question assigned to the following page: 3

3. KNOWN: Cross-section temperature profile of metal substrate under focused laser heating

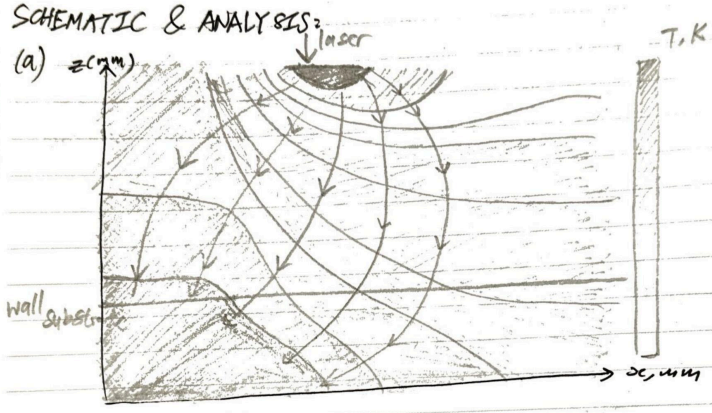
FIND: (a) Heat flux vectors

(b) Explain which side the laser flux has larger q''

ASSUMPTIONS: (1) 2D heat transfer within x-z plane

SCHEMATIC & ANALYSIS:

(a)



(b) The left side of the laser spot has larger heat flux because the isotherms are denser, indicating a higher rate of heat transfer.